

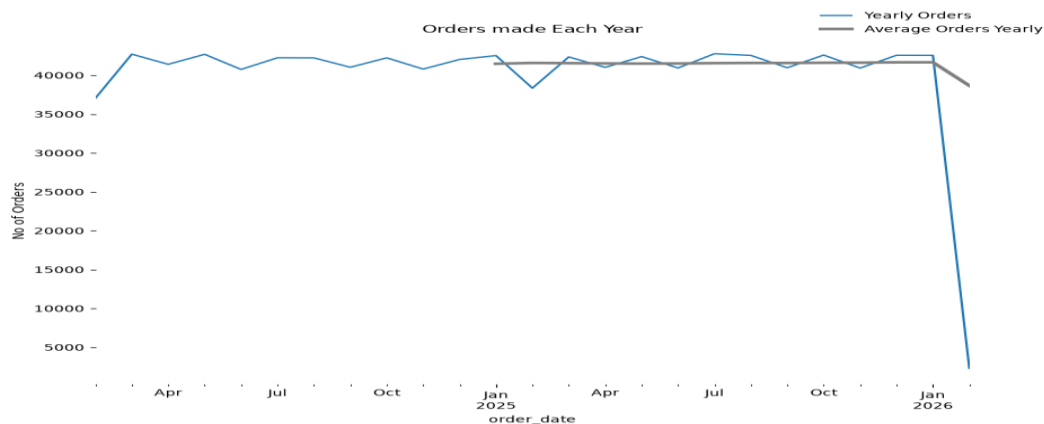
E-commerce Data Project

Steps

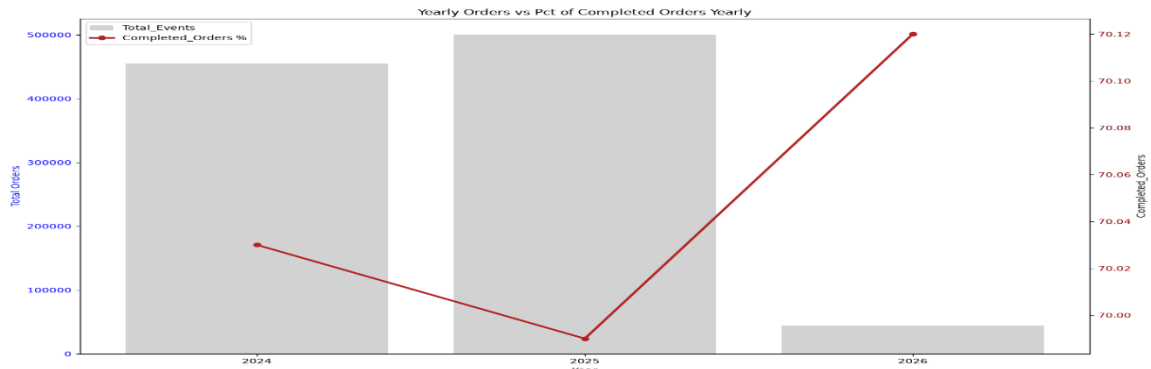
- Import libraries
- Load and read the dataset
- Quick Data Cleaning:
 - Return_reason column was dropped (its percentage of missing values was 90%)
 - Nan Values in coupon_code and customer_feedback was filled with no coupon and no feedback respectively
 - order_year data type int64 -> str
 - order_date data type str -> datetime[ns]

Insights

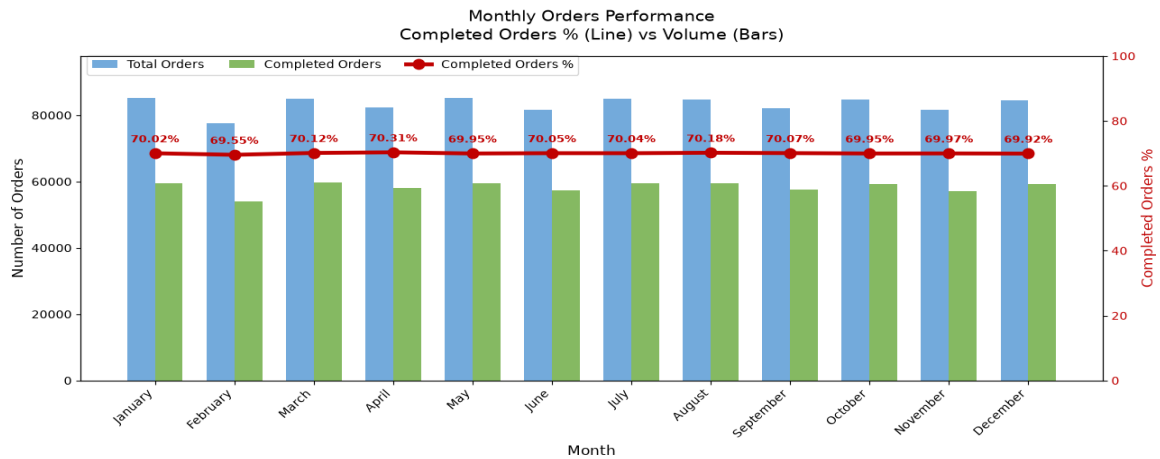
- From 2024 to 2025, total orders fluctuated consistently between 40000 and 43000 per month, maintaining a predictable cyclic/ seasonal cadence around an average line (Average Orders Yearly) of roughly 41,500 orders.
- The drop in sales at Feb 2026 is a result of incomplete data (no supporting data provided)



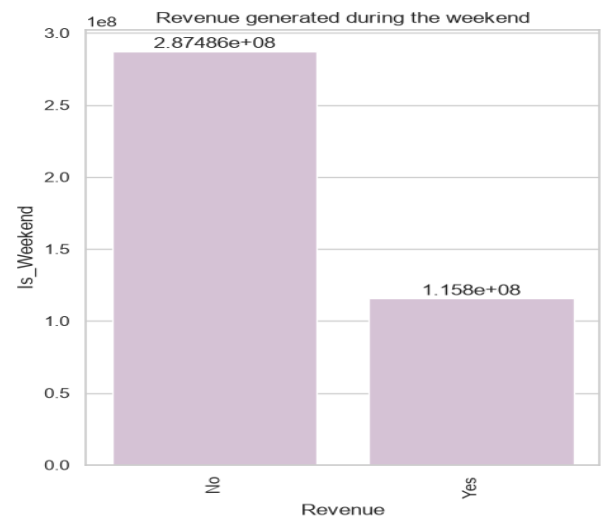
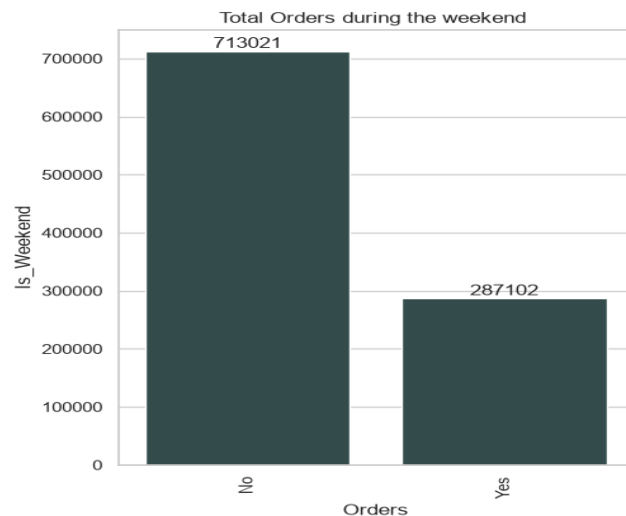
- Distribution of orders vs Pct of orders completed each year: figure shows that the firm maintains an operational consistency of 70%, unaffected by the scale variations or time periods.



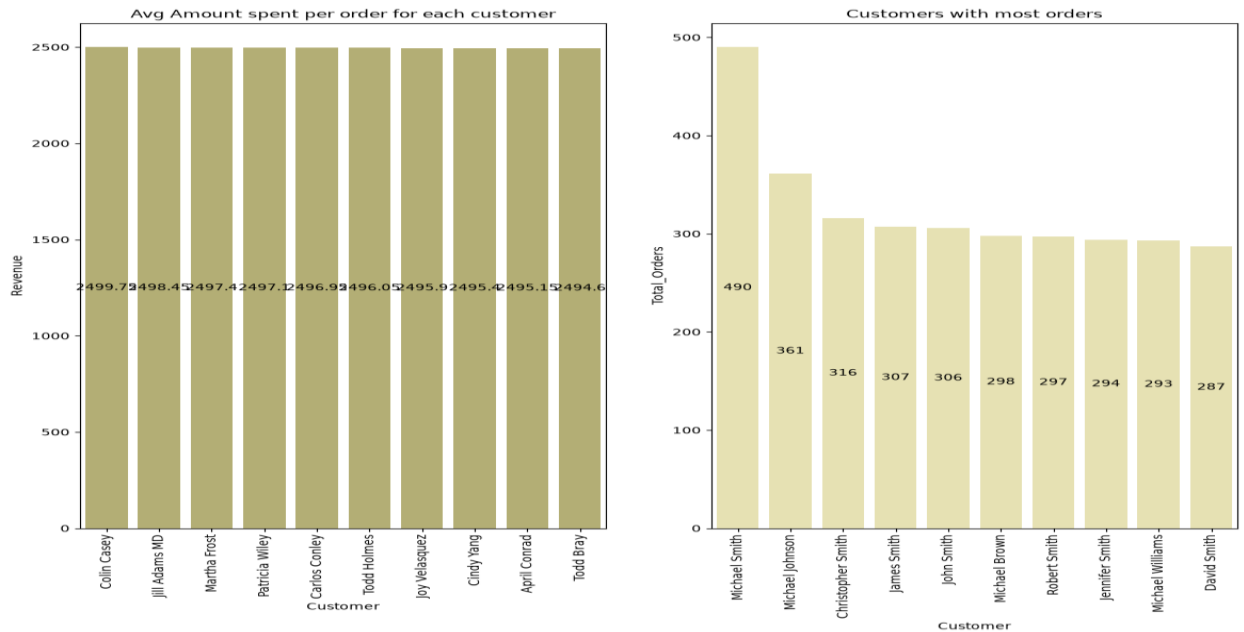
- Distribution of orders vs Pct of orders completed each month: figure still maintains what is being shown in the previous paragraph, fulfillment operates with extreme reliability regardless of monthly order volume fluctuations.



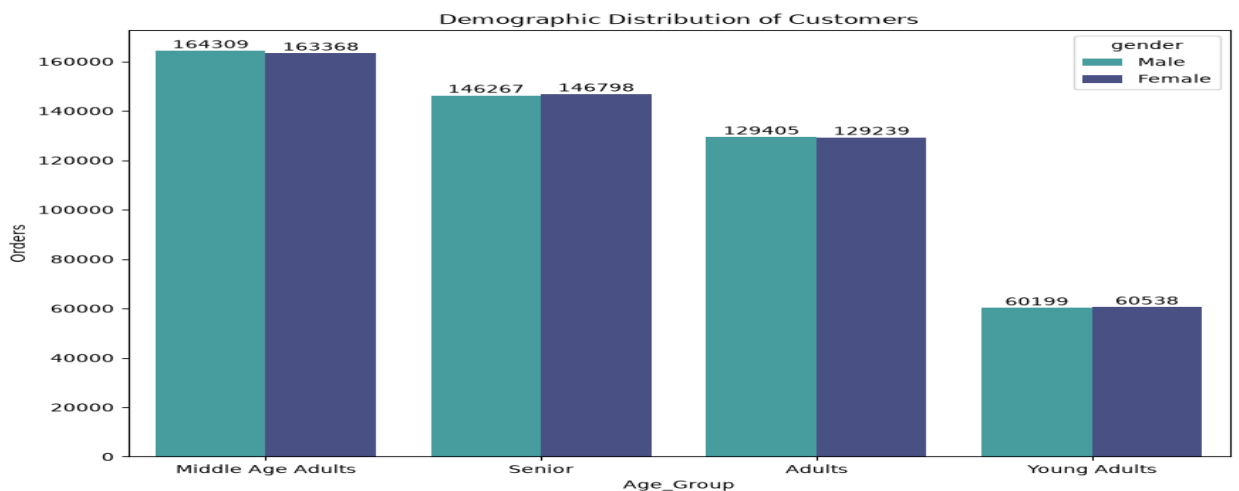
- Are there more orders/ sales made during weekends or weekdays: for both orders and revenue, weekdays take the largest share for both. Purchasing behavior is virtually identical on a per day basis, the higher aggregate weekday totals are a result of 5:2



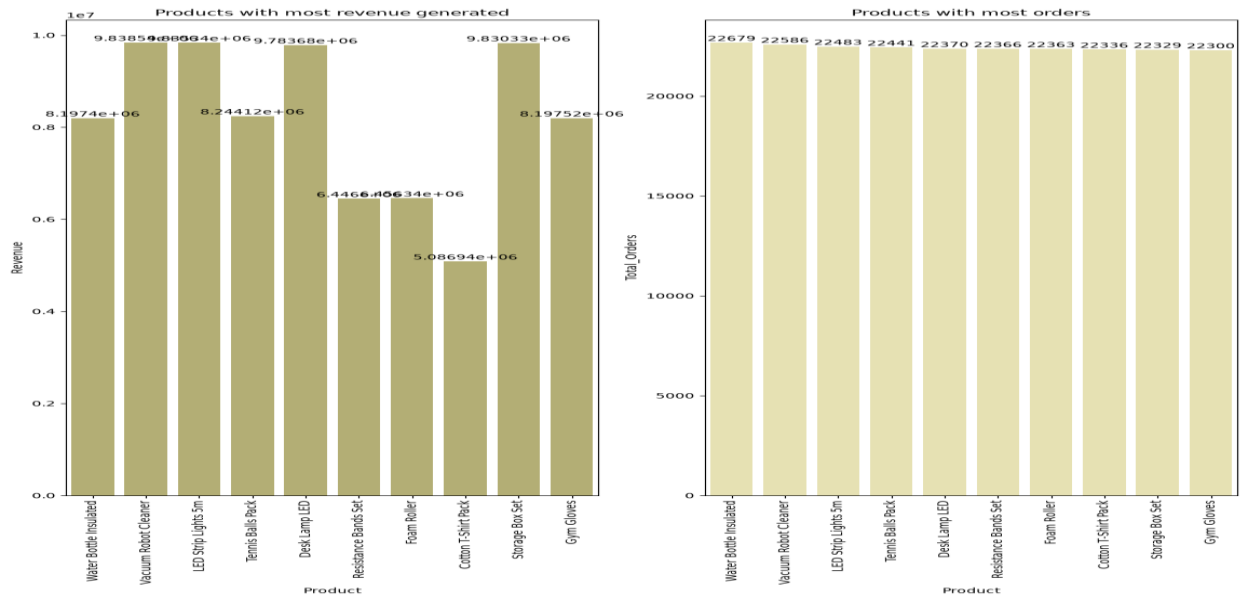
- Customers with most orders, average amount spent per order per customer: there is zero overlap between customers with high order volume and customers with high average order value. This near perfect clustering of aov around ~\$2500 across different top customers strongly point to a platform spending limit, a max price cap per basket or synthetic test data aggregation



- What is the demographic of customers: there is a near perfect split between male and female buyers in each age category. Gender has virtually no impact on purchasing volume.

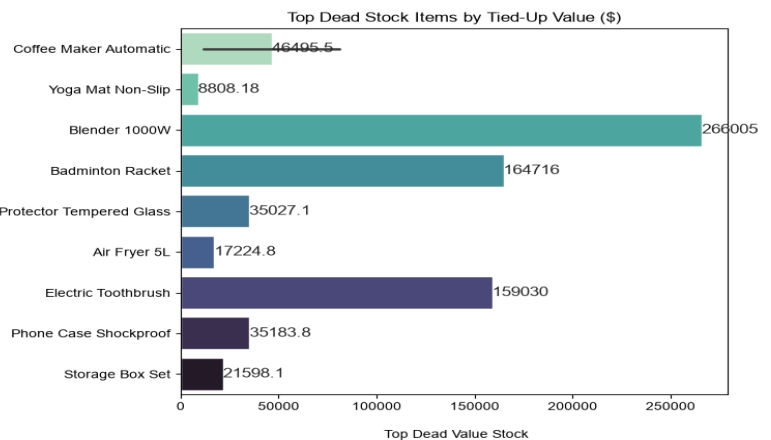
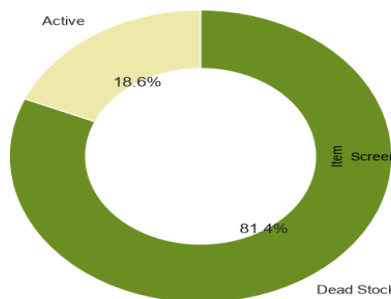


- Products with most sales volume/ high revenue generated: figure shows that differences in revenue generated is caused by variance in price. High order volume does not equal revenue generation.



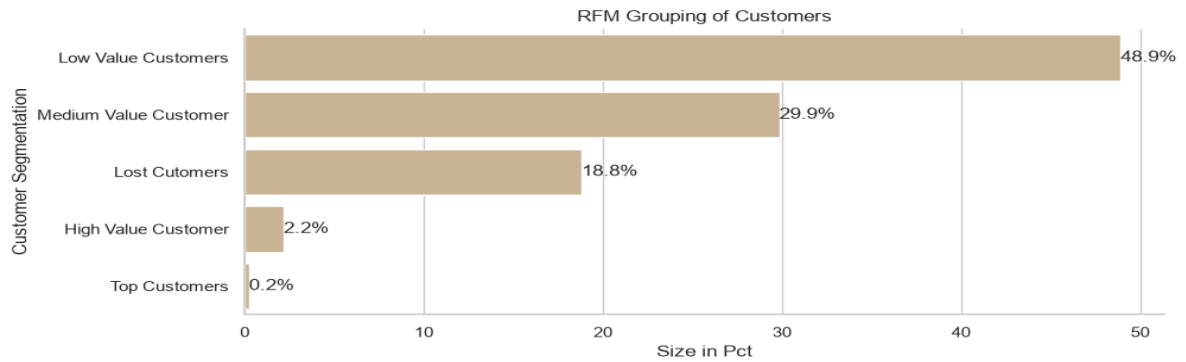
- Is there a presence of dead stock: more than 81% of total inventory is tied up in dead stock, it shows that working capital is heavily constrained by stagnant inventory, creating severe liquidity friction and increasing holding/warehousing costs.

Inventory Capital Allocation (Dead vs. Active)



Recommendations: the firm should run targeted discount campaigns, bundling promos or liquidations sales specifically for high value items like Blender 100W, Badminton Racket and electric toothbrush to convert dead stock into active cash flow. The firm should also reevaluate purchasing pipelines, demand forecasting models and MOQs to prevent over-stocking slow moving products.

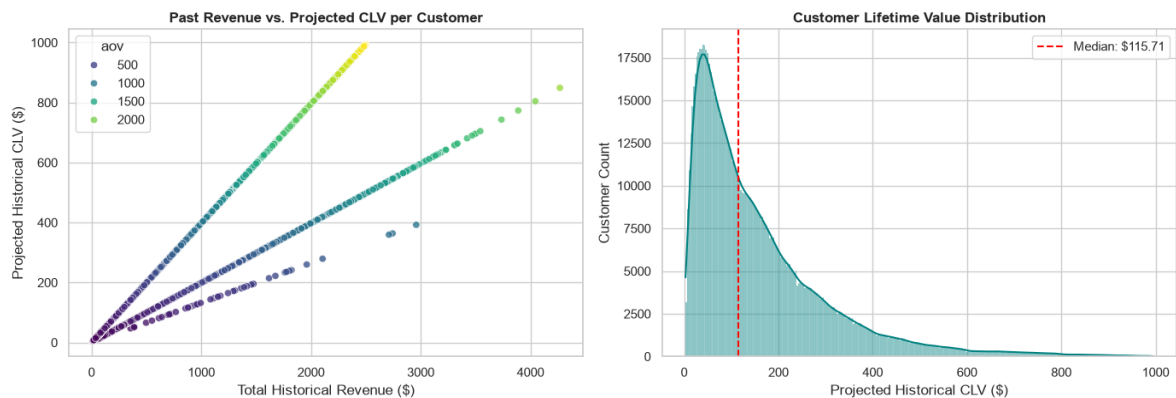
- What is the rfm segmentation like: it shows that the data is heavily skewed, the vast majority of accounts are casual or low frequency indicating a wide top of the funnel reach but significant untapped expansion potential.



It also shows that customers who purchased in large groups recently haven't repurchased, signaling weak post purchase engagements or lifecycle drop off.

Recommendations: Target 18.8% Lost Customers with win back offers, personalized product reminders or re-engagement discounts before they churn permanently.

- What is CLV like: The figure shows that vast majority of customers are concentrated in low-to-moderate CLV range, (\$0 to \$200), peaking around \$50-60 before tapering off.



Recommendation: since customers with higher average basket sizes convert past historical spend into projected lifetime value far more efficiently, marketing channels that attract higher spending demographics offer superior long-term ROI. Loyalty incentives, post purchase email flows should target low clv buyers since more than 50% of customer base belong there.